

Your finding, as a full sentence — the claim, not the topic

Resize and shift boxes to fit your visual elements

1 QUESTION

One line. What did you want to find out, and why would another scientist care?

~15 s

2 APPROACH — DRAW IT

~30 s

Schematic of the setup

*Show what you varied and what you held fixed.
A drawing beats a paragraph of method.*

one line of conditions: what changed, what stayed the same

3 RESULT — PLOT IT

~50 s

One figure

*Points with uncertainties, sized for the back row.
Show every trial; label the series on the plot.*

$n = \text{___}$ trials · bars = ___ · random or systematic?

4 TAKEAWAY → NEXT STEP

~25 s

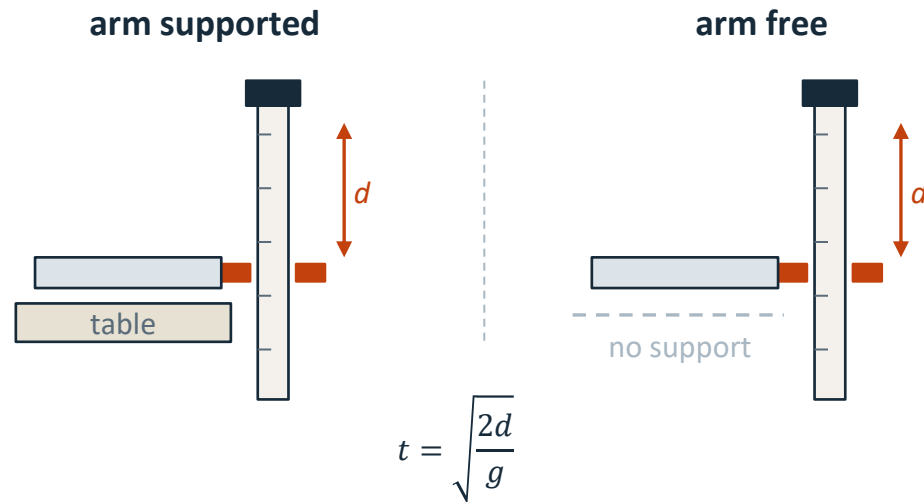
What it means, in one sentence · the next experiment · one honest limitation. Rehearse this line word for word.

Resting the forearm on the table shortens measured reaction time by 28 ± 11 ms

1 QUESTION

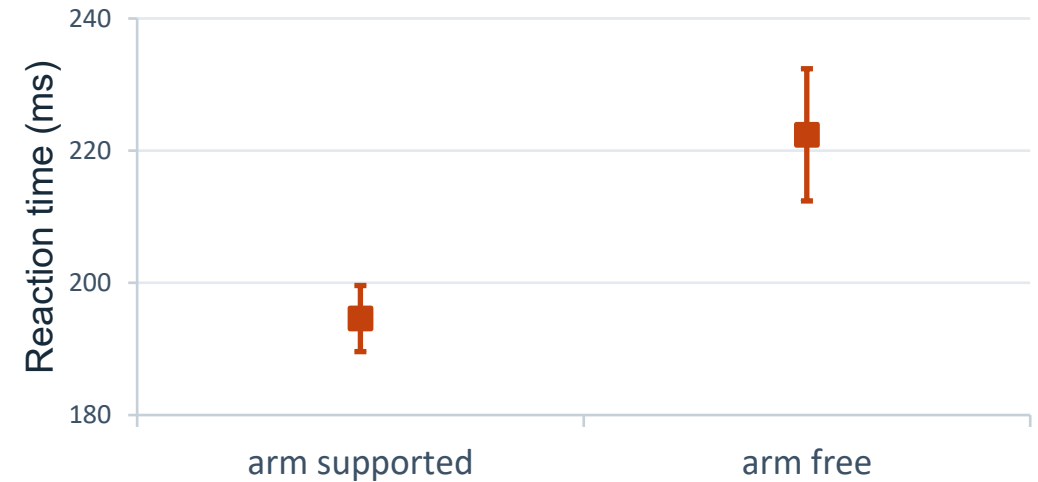
Does supporting your catching arm reduce the measured reaction time?

2 APPROACH



Same partner, same grip height. 5 catches per condition, order randomized.

3 RESULT



Points: mean $\pm$ propagated SEM, random only.

4 TAKEAWAY → NEXT STEP

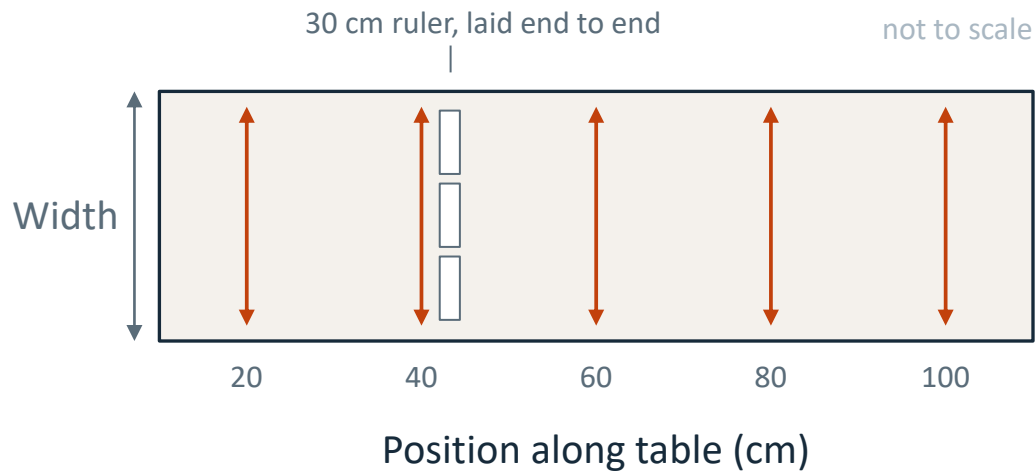
Unsupported arm causes a systematic shift and increases variation between trials.

Next step: hold posture fixed and swap who drops the stick. Limitations: one subject, one partner, and anticipation is uncontrolled.

Measured width varies 6.8 ± 2.0 mm along the table

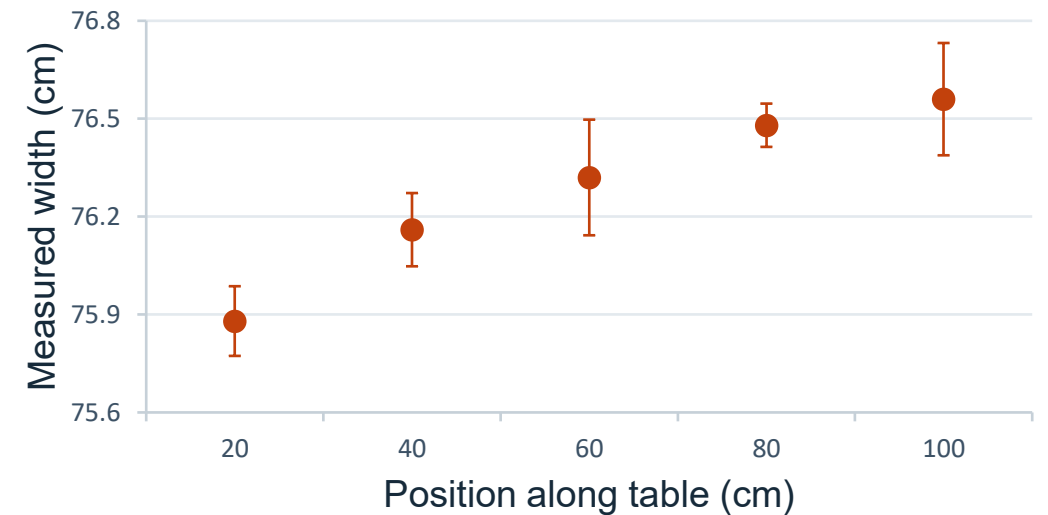
1 QUESTION Is the table we are measuring rectangular?

2 APPROACH



Width taken at 5 positions, 5 trials at each position. Trials randomized.

3 RESULT



Points: mean $\pm$ SEM. Difference between left and right edge is $7.6 \text{ mm} \pm 0.2 \text{ mm}$

4 TAKEAWAY → NEXT STEP

The difference in width between the opposite ends of the table is significant: **the table does not have a fixed width.**
The ruler path might have drifted which could make every measurement systematically too long.

One slide, two minutes maximum

Recommended: [Download PowerPoint for free](#) with your HMC email address and HMC Credentials password.

Guidance for creating the slide

The title is the result

State the main result in the title, not simply a topic. What do you want your audience to takeaway from your talk? Sometimes the systematic error you found is the result, not the number you got.

An Approach schematic should clearly show your procedure

A drawing answers “what did you do” faster than any sentence. The schematic should make the comparison obvious: what changed between conditions and what did you deliberately hold fixed? Schematics drawn either in PowerPoint (or other software) or something drawn neatly by hand can work well in Physics 50. Photos of a setup are usually less clear without taking significant extra care in setting up the photo to minimize visual distraction and then adding labels to the image.

Make the layout simple: we recommend a two-panel slide

The approach gets a schematic, the result gets a figure, side by side and about the same size. Make visual elements as large as possible without overcrowding the slide. Shrink the slide to 25% or stand two meters from your laptop to understand how your audience might see the slide. Check before you present: read your title out loud, then look only at your two pictures. A classmate/colleague should be able to get the point from those alone.

Your results should be displayed in a graph

Near any result, display number of trials and what the error bars represent. But don't add too much extra text. Use text for labels, rather than having long sentences. Words on the slide compete with your voice. If a line is not a title, a label or a caption, it belongs in your speaking notes.

Guidance for delivering the talk

Two minutes is about 260 spoken words. Below is a recommended, approximate time budget. Two thirds of your time goes on the two pictures. Say what is drawn before you say what it means — name the axes, the conditions, and what one data point is, then interpret.

15 s	30 s	50 s	25 s
Question	Approach — talk through the schematic	Result — talk through the figure	Takeaway + next

- Practice it out loud a few times and time it. A silent practice run typically is too fast.
- Separate observation from interpretation. “The width grew along the table” and “the table is a trapezoid” are different claims.
- Say plainly what you do not know. A clearly stated limitation reads as competence. Overclaiming is the fastest way to lose your audience.
- Don't be afraid to show a messy result. A systematic error you found but could not fully remove makes a better talk than a clean number. Say what you would do next.

Disclosure statement

To create this lightning talk template file, the Physics 50 teaching team fabricated experimental data and used generative AI to create the initial file. We iterated on that initial file in several subsequent revisions and instructor meetings to create the template for your use in Physics 50.

We acknowledge that it may seem unfair for us to use generative AI in developing this template to help you prepare for your oral report, while prohibiting you from using AI to assist you to analyze your data and understand your experiment. On careful reflection, we hope you will recognize that a key learning goal of this course is to teach you to analyze data and judge results carefully and that you will not develop this skill by outsourcing that judgment to AI. In providing you these templates, our goal is to give you a tool to focus on course content and not spend time on the formatting of your presentation.